

EPPS 6354 Information Management

Spring 2026

Thursday: 4:00 pm - 6:45 pm [GR3.206](#)

Instructor: Karl Ho; 972-883-2017; kyho@utdallas.edu | [GR3.203](#)
Office Hours: Monday, Wednesday, 1:30 pm – 3:30 pm, by appointment.
Teaching Assistant: Ariel Wang | Ariel.Wang@UTDallas.edu | [GR 3.318](#)
Office Hours: Monday, Wednesday 1:30 pm to 3:30 pm, by appointment.

Overview:

This is an application course introducing database systems including the concepts, design, programming and management of database systems. This course equips students with fundamental knowledge of database, applications and practical skills to design, create and manage a database system. Class materials will cover not only traditional relational database but introduce new technologies such as NoSQL, **vector databases**, and various types of modern database systems **including AI-database integration technologies**. Students will be heavily engaged in database programming and building applications. At end of the course, students should be able to understand extensive database system concepts, **critically evaluate AI-generated database solutions**, and build a research database system.

Prerequisites:

There is no official class requirement before taking this course. The first prerequisite is willingness to learn the materials that could be out of your current field/discipline. The second is the strong desire to build and scale using the materials covered in class.

Learning Objectives:

At completion of this course, students will be able to:

1. Understand relational database concepts
2. Manipulate data in Database Management Systems (DBMSs)
3. Be fluent in the SQL language
4. Design database applications
5. Be familiar with modern database technologies
6. **Critically evaluate AI-generated SQL queries and database designs**
7. **Understand vector databases and retrieval-augmented generation (RAG) architectures**
8. **Apply data governance and quality principles to research contexts**

Required Text:

Silberschatz, Abraham, Korth, Henry F. and Sudarshan, S., 2019. *Database system concepts*, 7th edition. New York: McGraw-Hill.
Foster, Ian, Rayid Ghani, Ron S. Jarmine, Frauke Kreuter, and Julia Lane, eds. 2020. *Big data and social science: Data science methods and tools for research and practice*. CRC Press.
Molino, Anthony. 2005. *SQL Cookbook: Query Solutions and Techniques for Database Developers*. O'Reilly Media, Inc. (‡)
Christof Strauch. 2009. NoSQL databases. <https://www.christof-strauch.de/nosql dbs.pdf>
Wickham, Hadley, Mine Cetinkaya-Rundel & Garrett Grolemund. Second edition. 2023. *R for Data Science*. Latest online edition (<https://r4ds.hadley.nz>)
Hellerstein, Joseph M, Vikram Sreekanti, Joseph E Gonzalez, et al. 2017. "Ground: A Data Context Service." (<https://rise.cs.berkeley.edu/wp-content/uploads/2017/03/CIDR17.pdf>)
Abadi, Daniel, Anastasia Ailamaki, David Andersen, et al. 2022. "The Seattle Report on Database Research." *Commun. ACM* 65(8): 72–79. doi:[10.1145/3524284](https://doi.org/10.1145/3524284).
Castro Fernandez, Raul, Aaron J. Elmore, Michael J. Franklin, Sanjay Krishnan, and Chenhao Tan. 2023. "How Large Language Models Will Disrupt Data Management." *Proc. VLDB Endowment* 16(11): 3302-3309. ([ACM Digital Library](#))

Zhou, X., Sun, Z. & Li, G. 2024. "DB-GPT: Large Language Model Meets Database." *Data Science and Engineering* 9: 102-111. ([Springer](#))
Pina, Eduardo Manuel, Filipe Alexandre Sá, and Jorge Bernardino. 2023. "NewSQL Databases Assessment: CockroachDB, MariaDB Xpand, and VoltDB." *Future Internet* 15(1): 10. ([MDPI Open Access](#))
Wang, Zige, et al. 2024. "Data Management For Training Large Language Models: A Survey." *arXiv:2312.01700*. ([arXiv](#))

(‡) Available electronically at McDermott Library

Textbook website:

<https://www.db-book.com>

Recommended Reading:

Raghu Ramakrishnan and Johannes Gehrke. *Database Management Systems*, 3rd Edition. McGraw-Hill, 2002.
Daniel Abadi, Rakesh Agrawal, et al. 2014. The Beckman Report on Database Research. *ACM SIGMOD Record*, 43(3):61–70, <http://beckman.cs.wisc.edu/beckman-report2013.pdf>.
Abadi, D., et al., 2016. The Beckman report on database research. *Communications of the ACM*, 59(2), pp.92-99.
Pokorny, J., 2013. NoSQL databases: a step to database scalability in web environment. *International Journal of Web Information Systems*, 9(1), pp.69-82.
Indrawan-Santiago, M., 2012. Database research: Are we at a crossroad? Reflection on NoSQL. In *2012 15th International Conference on Network-Based Information Systems* (pp. 45-51). IEEE.
CMU 15-445/645 Database Systems Course Materials (<https://15445.courses.cs.cmu.edu>)
Pinecone Learning Center: Vector Database Guide (<https://www.pinecone.io/learn/vector-database/>)

Grading and Requirements:

Personal website:

Each student is required to create and design a website hosted on GitHub. All assignments and class materials will be posted on own website before next class. Consult Building Personal Website on GitHub online workshop to create the GitHub website. Using the Quarto method is required. Personal GitHub website is responsible for **15** percent of the final grade.

Participation:

Full attendance of all classes is required and imperative. Attendance however is not enough. All class members have to actively participate in class preparation and discussion. Participation entails full preparation for class including intensive research of class materials, completion of assigned readings and full involvement in class discussion. Only medical emergency (with documentation) is considered excused absence. If you have a special medical condition, please contact me in first week of semester and special arrangements can be made. Participation is responsible for 10% of the final grade.

Assignments:

Exercises will be assigned. Students are expected to finish all assignments and apply techniques to own project. For example, techniques in database design, charts and applications should be applied in proposal, progress presentation and final project and presentation. At beginning of each class, one student will be randomly selected to present and discuss assignment solutions. Assignments (posted on website) are responsible for 10 percent of the final grade.

AI-Assisted Learning Labs:

Three structured lab assignments where students work with AI tools to deepen understanding of database concepts. These labs develop critical evaluation skills essential in the AI era:

- **Lab 1: Query Evaluation** – Use AI to generate SQL queries, then analyze execution plans and identify optimization opportunities the AI missed.
- **Lab 2: Schema Design Critique** – Compare three AI-generated schema designs for a given problem, justify your choice, and improve the selected design.
- **Lab 3: RAG Implementation** – Build a simple retrieval-augmented generation system using vector embeddings and a research corpus.

AI-Assisted Learning Labs are responsible for **15 percent** of the final grade.

Database Project:

Each student must design and implement a database project using the systems and techniques covered in class. Identify one original topic for database system. Original means the database must be originally designed and created. Data can be collected from existing source. The final report on design and implementation of database system must be between 10-20 pages in length. Originality is the first quality the instructor demands. Due dates of the proposal presentations are listed on the schedule below. No late submission will be accepted. Proposal of the project must be consulted with the instructor and subject to approval. The proposal and progress presentations account for **15%** of final grade. The final report, with in-class presentation, is responsible for **35** percent of the final grade.

In summary, the grade structure is as follows:

Personal GitHub website:	15%
Participation:	10%
Assignment (posted on website):	10%
AI-Assisted Learning Labs:	15%
Project proposal and progress presentations:	15%
Final project and presentation:	35%

Document guideline:

All documents in this class must adhere to the following general guidelines:

- Submit electronic document to instructors via email in PDF format (NO Word or GoogleDoc):
- With author name and content on file name: E.g. smith_proposal.pdf, doe_progressreport1.pdf
- on US letter size paper, one inch on all margins, 12-point font
- NO submission using link to cloud or online drive (e.g. Google drive, box.com)
- Following the APSA style and format is mandatory (APSA style manual (2018 edition): <https://connect.apsanet.org/stylemanual/>).

Tentative Schedule (This schedule is subject to change):

Week	Topic	Reading	Due
1	Introduction	IF 1, 4	
2	Relational Languages	SKS 1, 2	
3	SQL I + AI Query Tools	SKS 3, 4; LLM-DB paper	
4	SQL II	SKS 5	AI Lab 1 [NEW]
5	ERM and Relational Database Design	SKS 6, 7	
6	Complex Data Types (JSON, Arrays, Vectors)	SKS 8; Vector DB Guide	Proposal (2/26)
7	Application Design + AI-DB Integration	SKS 9; DB-GPT paper	AI Lab 2 [NEW]
8	Big Data	SKS 10; IF 5	
9	<i>Spring break</i>		
10	Data Analytics	SKS 11	Progress 1 (3/26)
11	Vector Databases & RAG Architectures	LLM Data Mgmt Survey; Seattle Report	
12	Query Processing	SKS 15	AI Lab 3
13	Database-System Architectures	SKS 20; NewSQL paper	Progress 2 (4/16)
14	Parallel/Distributed Storage + Data Governance	SKS 21, 22; Ground paper	
15	New Developments in Data/Information systems	SKS 28, 29, 30, 32; Beckman reports	
16	Final Presentations		Final Project (5/7)

Yellow highlighted rows indicate new or significantly updated content.

Workshop Topics:

GitHub Website with Quarto
PostgreSQL
Dashboard
Shiny
(proposed)
pgvector (PostgreSQL extension for vector similarity)
DuckDB (embedded analytical database)

Comet Creed

This creed was voted on by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:

"As a Comet, I pledge honesty, integrity, and service in all that I do."

Generative AI policy

*Generative AI is **explicitly integrated into this course** as both a subject of study and a learning tool*

In this course, students shall give credit to AI tools whenever used, even if only to generate ideas rather than usable text or illustrations. When using AI tools on assignments, add an appendix showing (a) the entire exchange, highlighting the most relevant sections; (b) a description of precisely which AI tools were used (e.g. ChatGPT private subscription version or DALL-E free version), (c) an explanation of how the AI tools were used (e.g. to generate ideas, turns of phrase, elements of text, long stretches of text, lines of argument, pieces of evidence, maps of the conceptual territory, illustrations of key concepts, etc.); (d) an account of why AI tools were used (e.g. to save time, to surmount writer's block, to stimulate thinking, to handle mounting stress, to clarify prose, to translate text, to experiment for fun, etc.).

For AI-Assisted Learning Labs specifically, students should also document: (e) what modifications were made to AI-generated output, and (f) reasoning for those modifications. The goal is not to avoid AI but to use it wisely and reflectively while developing the expertise to recognize when AI suggestions are incorrect or suboptimal.

Students shall not use AI tools during in-class examinations, or assignments unless explicitly permitted and instructed. Overall, AI tools should be used wisely and reflectively with an aim to deepen understanding of subject matter.

(adapted from the University of Texas at Austin Center for Teaching and Learning ChatGPT and Generative AI Tools: Sample Syllabus Policy Statements)

Academic Support Resources

The information contained in the following link lists the University's academic support resources for all students. Please see <http://go.utdallas.edu/academic-support-resources>.

UT Dallas Syllabus Policies and Procedures

The information contained in the following link constitutes the University's policies and procedures segment of the course syllabus. Please review the catalog sections regarding the credit/no credit or pass/fail grading option and withdrawal from class.

Please go to <http://go.utdallas.edu/syllabus-policies> for these policies.

The descriptions and timelines contained in this syllabus are subject to change at the discretion of the Professor.